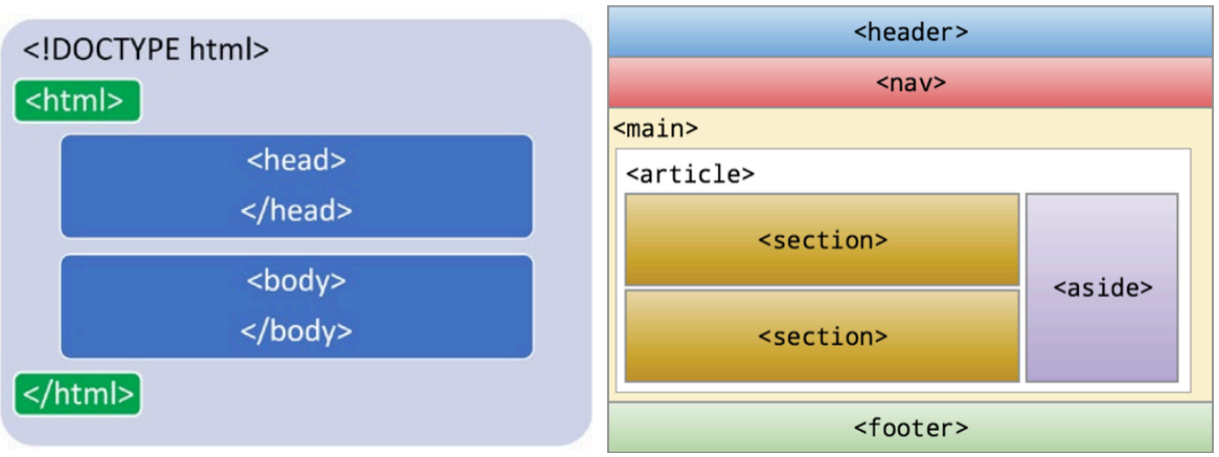
Modul 1 : HTML Basic



Structure of an HTML Document

- HTML stands for Hyper Text Markup Language
- HTML is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

1. HTML Elements

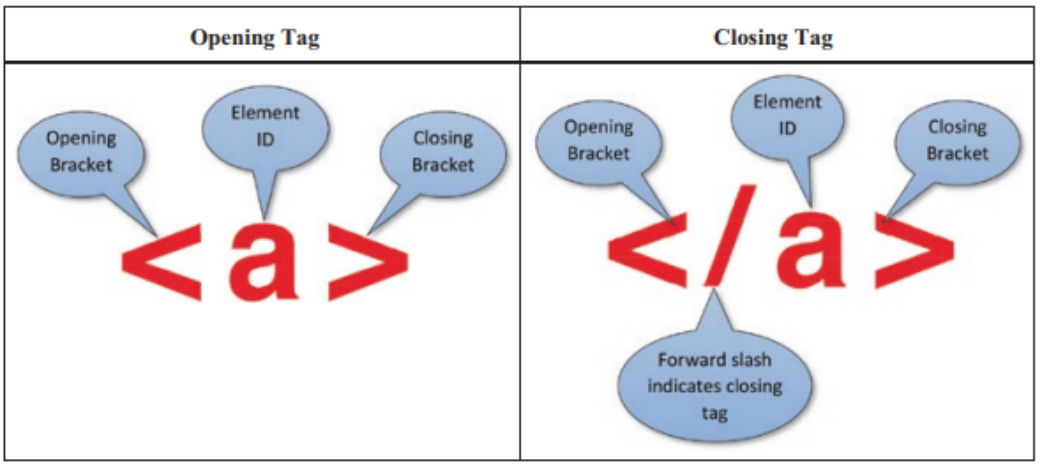
The HTML element is everything from the start tag to the end tag:

```
<tagname>Content goes here...</tagname>
```

Examples of some HTML elements:

```
<h1>My First Heading</h1>
<p>My first paragraph.</p>
```

2. HTML ELement Structure



2.1. HTML Opening and Closing Tags

The opening HTML tag often contains some attributes that define the HTML element's properties and are used to control formatting, size, page link references, and so on, and it is placed inside the element's opening tag. For example

HTML **attributes** are made up of two parts: **a name** and **a value**.

- The name is the attribute you want to set. For example, the anchor element contains an attribute named "href," which indicates the address of the page you want to link to.
- The value is what you want the attribute to be set to and is always contained within quotation marks. In this example, the page we want to link to is called "about.html"

2.2. Nested HTML Elements

HTML elements can be nested (this means that elements can contain other elements). All HTML documents consist of nested HTML elements.

```
<body>
     <h1>My First Heading</h1>
     <p>My first paragraph.</p>
</body>
```

3. Metadata

Metadata is additional information about an HTML document. The meta elements can be used to describe properties of the HTML document, such as author, date, and content descriptions. Metadata is used by browsers to determine how to display content and by search engines such as Bing or Google to work out what a web page is about.

`<meta>` always go inside the `<head>` of the HTML page as you can see in the following:

```
<head>
    <meta charset="UTF-8">
    <meta name="description" content="Enjoy succulent meats…">
    <meta name="author" content="Anna Wilson">
</head>
```

The meta element has various attributes such as charset and name. A meta charset specifies the character encoding for the HTML document which is usually UTF-8.

A meta name specifies names for the metadata such as a content description that appears in search engine results of the page, keywords that identify the content, and author of the page.

4. Creating Links

The HTML tags that do the link thing are <a> and </a>. Here's how the <a> tag works:

```
<a href="address">
```

`href` stands for hypertext reference, which is just a fancyschmancy way of saying "address" or "URL." Your job is to replace address with the actual address of the web page you want to use for the link. And yes, you have to enclose the address in quotation marks (“). The form of address value you use depends on where the web page is located with respect to the page that has the link.

There are three possibilities:

4.1. Remote Web Page :

Refers to a web page that's not part of your website. In this case, the <a> tag's href value is the full URL of the page.

```
<a href="https://vokasi.unair.ac.id">
```

4.2. Local web page in the same directory :

Refers to a web page that's part of your website and is stored in the same directory as the HTML file that has the link. In this case, the <a> tag's href value is the filename of the page.

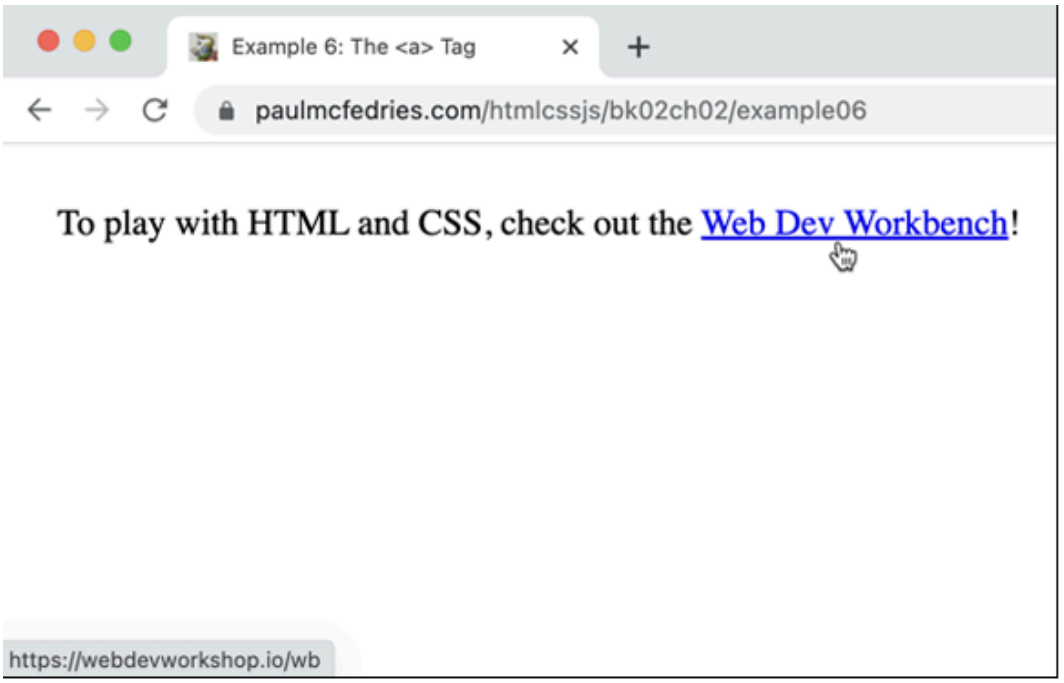
```
<a href="Tentangkami.html">
```

4.3. Local web page in a different directory :

Refers to a web page that's part of your website and is stored in a directory other than the one used by the HTML file that has the link. In this case, the tag's href value is a backslash (/), followed by the directory name, another backslash, and then the filename of the page.

```
<a href="/admin/dashboard.html">
```

Descriptive Link




```
To play with HTML and CSS, check out the
<a href="https://webdevworkshop.io/wb">
Web Dev Workbench</a>!
```

Anchors aweigh: Internal links

Internal links actually link to a specially marked section — called an anchor — that you've inserted somewhere in the same page. To understand how anchors work, think of how you may mark a spot in a book you're reading.

An anchor tag looks a lot like a regular tag, except that it also includes the `id` attribute, which is set to the name you want to give the anchor. Here's an example:

```
<section id="section1">
```

You can use whatever you want for the name, but it must begin with a letter and can include any combination of letters, numbers, underscores (_), and hyphens (-). Also, id values are case-sensitive, so the browser treats the id value section1 differently than the id value Section1. To set up the anchor link, you create a regular <a> tag, but the href value becomes the name of the anchor, preceded by a hash symbol (#):

```
<a href="#section1">
```

5. Adding Media

5.1. Inserting Image

HTML code that tells a browser to display an image is the `<img>` tag, and here's how it works:

```
<img src="filename" alt="description" title="title">
```

`src` is short for source; `filename` is the name (and often also the location) of the graphics file you want to display; `description` is a short description of the image (which is read by screen readers and seen by users who aren't displaying images or if the image fails to load); and `title` is a tooltip that appears when the user hovers the mouse pointer over the image.

If you have a logo, icon, or other picture that you want your readers to click to go to a different page, you can set up that image as a link.

Here's the general technique to use:

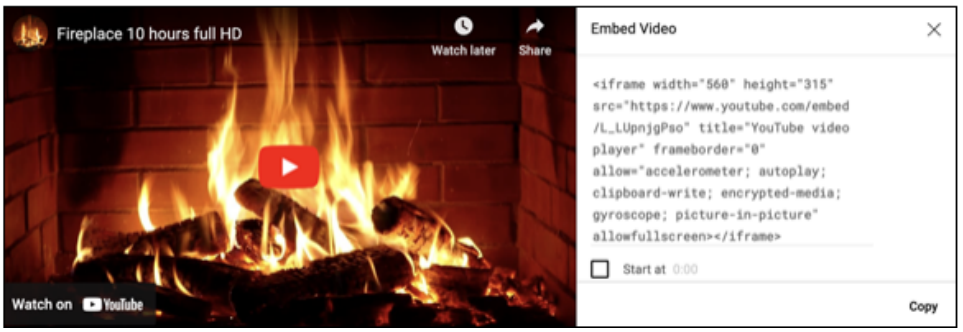
```
<a href="address"><img src="filename"
alt="description"></a>
```

5.2. Embedding Videos

How you embed a video into a web page depends on the type of video:

- **Third-party video:** For videos from third-party sites such ascYouTube, Vimeo, and Dailymotion, you insert some HTML codecinto your page that automatically plays the video from the third-partycSite.

  YouTube: Go to youtube.com and find the video you want to embed in your page. Click the Share button. In the Share dialog box that appears, click Embed to display the code



- **Your own video:** For video files that you create yourself, you upload the file to your server and then use the video element to insert the video into your page.

How you use the video element depends on whether you're offering a single video file format or multiple formats.

```
<video src="source"
 poster="poster"
 width="width"
 height="height"
 autoplay
 controls
 loop
</video>
```

`src:` The name of the video file, if the file is in the same location as the HTML file. If the video file resides in a different directory, you need to include that info so that the browser knows where to look for the file.

`poster:` The name (and directory, if needed) of an image that you want the browser to display before video playback begins. Most of the time you use a title frame or still frame from the video.

`width:` The width, in pixels, of the video playback window.

`height:` The height, in pixels, of the video playback window.

`autoplay:` Include this attribute to have the browser automatically start playing the video as soon as enough of the file has been downloaded for smooth playback. If you don't want the video to start automatically, omit the autoplay attribute.

`controls:` Include this attribute to have the browser display the playback controls in the video window. If you don't want your visitors to be able to control the playback, omit the controls attribute.

`loop:` Include this attribute to have the browser automatically begin playback from the beginning whenever the video ends. If you want the video to play only once, omit the loop attribute

If you want to offer multiple video formats, remove the src attribute from the <video> tag and add multiple source elements, one for each format you want to offer:

```
<video
      poster="filename"
      width="value"
      height="value"
      autoplay
      controls
      loop>
      <source src="filename"
            type='type; codecs="codecs"'>
       <source src="filename2"
            type='type; codecs="codecs"'>
      <source src="filename3"
            type='type; codecs="codecs"'>
      <source etc….
</video>
```

Exercises

Create a single HTML page consisting of 4 sections:

1. Section 1: Header Banner

   Display a banner photo (size and orientation are up to you, but landscape orientation is recommended).

2. Section 2: Text Paragraph with Hyperlink

   Include a paragraph of text along with a hyperlink to an external website.

3. Section 3: Image Gallery with Links

   Create a gallery of images with each image linked to a different webpage or resource.

4. Section 4: Embedded YouTube Video

   Embed a YouTube video of your choice into this section.

Instructions:

- Design a basic HTML page with the specified sections.
- Ensure that each section is appropriately labeled using HTML tags.
- Test your page in a web browser to ensure all elements display correctly.

Feel free to be creative with the content and layout of each section. This exercise will help reinforce your understanding of HTML fundamentals and embedding various types of content into web pages.

Name your file : **NIM_NAMA_Modul1.html**